

POMDP-Based Target Tracking in Crowded Human Environment

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Abstract—This project addresses the problem of tracking a moving target in a dynamically crowded human environment. Safety is treated as a primary objective. The robot must reliably avoid collisions with nearby humans while simultaneously pursuing the target. The problem is formulated as a Partially Observable Markov Decision Process (POMDP), and several planning approaches are explored for robust target tracking under limited sensing. To handle uncertainty in the target’s state, a particle filter is used to maintain a probabilistic belief over the moving target in the environment. The planner uses this belief to predict where the target is likely to go and generates interception strategies that balance tracking performance and safety. An online tree search algorithm (DESPOT) is implemented to select actions that explicitly trade off tracking performance and safety. Experiments show that using DESPOT to control the robot’s heading and speed leads to improved tracking accuracy and safer navigation compared to baseline planners. Overall, the integration of particle filtering with online tree-search planning provides a promising solution for target tracking in dynamic, cluttered environments where safety is critical.

Index Terms—POMDP, Belief, Particle Filter, Target Tracking, Interception, Hybrid A*

I. INTRODUCTION

At the core of many robotics and autonomous systems applications lies the challenge of making safe and informed decisions in uncertain, dynamic environments. A particularly critical instance of this challenge is navigating safely through dense crowds of moving humans. This problem is inherently difficult in that the robot must continuously perceive and predict the motion of nearby humans while simultaneously planning its own trajectory to avoid collisions and reach its objective in a timely manner. The complexity is further amplified by uncertainty in human behavior, sensor noise, and the additional requirement of tagging an uncooperative, dynamically moving target in the environment. These uncertainties necessitate probabilistic reasoning and decision-making frameworks that can plan effectively under incomplete and noisy information. Efficiently addressing this problem has significant implications for surveillance and reconnaissance, search-and-rescue operations, and service robotics in crowded public spaces.

This work builds on the safe autonomous navigation framework in crowded human environments proposed by Gupta et al. [1], where the task was to reach a static goal location while avoiding collisions. Their formulation demonstrated the

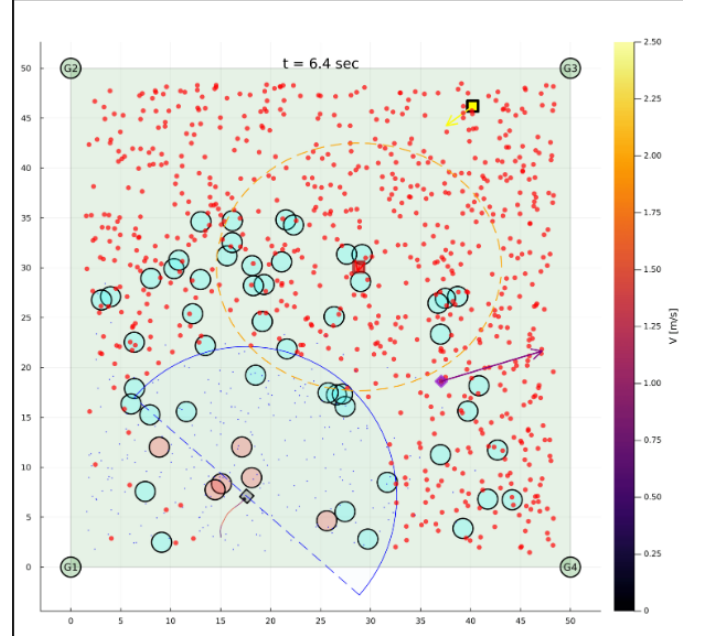


Fig. 1. Snapshot of the simulation environment used to evaluate the planners. The autonomous vehicle (blue square) is equipped with a LiDAR sensor whose finite field of view is shown as a semicircular arc. Its task is to track and tag a moving target (yellow square) while avoiding collisions with humans (blue circles) in a crowded environment. Among these, the six nearby humans that are explicitly modeled in the POMDP solver are highlighted as orange circles. The red particles represent samples from the belief over the target’s state; the red square with an uncertainty ellipse denotes the mean of this belief; and the purple square denotes the MAP estimate. Arrows attached to the target, belief mean, and MAP estimate indicate their current velocity direction.

potential of using online tree search to jointly control the robot’s heading and speed for improved safety and efficiency. Building upon this foundation, this project extends the problem from static goal navigation to dynamic target tracking in crowded human environments with a non-holonomic robot. The objective is to demonstrate that online tree-search methods such as DESPOT can not only safely drive the robot toward a fixed goal, but also be employed to anticipate and intercept a moving target while avoiding collisions with nearby humans.

In this work, three planning approaches are compared within the same problem setting. The first is a baseline greedy planner that uses hybrid A* to compute a path toward the most likely

current location of the target, given the belief, and then uses DESPOT to control the robot's speed along this path while avoiding collisions. The second is an anticipatory planner that uses hybrid A* to plan a path toward the predicted future location of the most likely target estimate, based on its current belief and heading, again using DESPOT to regulate the robot's speed along this path under safety constraints. The third approach is a POMDP planner that directly uses DESPOT to control both heading and speed, allowing the robot to balance tracking performance and safety. The performance of these planners is evaluated using metrics such as success rate, safety (measured by the number of collisions), and efficiency (measured by the time required to intercept the target).

The remainder of this report is organized as follows. Section II reviews related work on dynamic target tracking and safe navigation in human-crowded environments. Section III presents the problem formulation. Section IV describes the solution approach based on DESPOT and the different planner variants. Section V reports the experimental results and analysis.

II. BACKGROUND AND RELATED WORK

Hybrid A* has been widely adopted in autonomous vehicle navigation due to its ability to generate kinodynamically feasible paths that respect the vehicle's constraints [2]. However, traditional deterministic path-planning algorithms do not explicitly perform probabilistic reasoning about uncertainty in dynamic environments.

To address this limitation, Partially Observable Markov Decision Processes (POMDPs) provide a principled framework for planning under uncertainty. They enable the robot to maintain a belief over partially observed aspects of the environment and to choose actions that maximize expected long-term reward. Earlier work by Bai et al. [3] proposed a two-stage planning framework that uses hybrid A* to compute a reference path and a POMDP to optimize speed control under uncertainty. While this approach demonstrated improved safety when navigating near pedestrians, it restricted the POMDP to a limited subset of the control space (longitudinal speed along a fixed path), often causing the robot to stop when the reference path was blocked. Gupta et al. [1] extended this idea by using a POMDP to jointly optimize both heading and speed, while leveraging multi-query motion-planning techniques to generate effective rollout policies that guide the online tree search.

Existing works, however, largely focus on individual components of the broader problem. Many approaches address autonomous navigation to a static goal in a dynamic environment [4], while others focus on tracking a target in a largely static environment, often formulated as a pursuit–evasion game [5]. In contrast, the challenge of simultaneously tracking a moving target while avoiding collisions with dynamic obstacles such as moving pedestrians, has received relatively less attention. This project aims to help bridge this gap by proposing a POMDP-based framework that integrates target tracking and collision avoidance in crowded human environments.

III. PROBLEM FORMULATION

A. POMDP Preliminaries

Partially Observable Markov Decision Processes (POMDPs) provide a mathematical framework for modeling sequential decision-making problems in which an agent must act under uncertainty about the true state of the environment. Formally, a POMDP is defined by the tuple $(S, A, T, R, O, Z, \gamma)$. At each time step, the environment is in some (unobserved) state $s \in S$. The agent chooses an action $a \in A$, and the environment transitions to a new state $s' \in S$ according to the transition function $T(s, a, s') = P(s'|s, a)$. The agent then receives an observation $o \in O$, generated according to the observation function $Z(s', a, o) = P(o|s', a)$, and obtains a reward $R(s, a)$. The objective is to find a policy π that, given the agent's information, maximizes the expected discounted cumulative reward starting from a uniform initial belief b

$$V^\pi(b) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \mid b, \pi \right], \quad 0 \leq \gamma < 1, \quad (1)$$

where $R(s_t, a_t)$ is the immediate reward received at time t and $\gamma \in (0, 1)$ is the discount factor.

$$\pi^* = \arg \max_{\pi} V^\pi(b). \quad (2)$$

Because the agent cannot directly observe the true state s , it maintains a belief state b , which is a probability distribution over all possible states S . After taking action a and receiving observation o , the belief is updated using Bayes' rule:

$$b'(s') = \eta Z(s', a, o) \sum_{s \in S} T(s, a, s') b(s), \quad (3)$$

where η is a normalizing constant chosen so that $\sum_{s' \in S} b'(s') = 1$.

B. POMDP Modeling

In this project, two variants of the POMDP planner are used. The Extended Space (ES) planner controls both the steering and speed of the vehicle, while the Limited Space (LS) planner only controls the vehicle's speed along a precomputed path (i.e., from the hybrid A* planner), with steering fixed by that path. The POMDP model for this problem is defined as follows:

- **States (S):** The state space includes the vehicle's pose and speed, (x, y, θ, v) , the state of each nearby human, represented by its position, speed, and sampled goal location, (h_x, h_y, h_v, h_g) , and the target position and speed (t_x, t_y, t_v) . Together, these components describe the dynamic configuration of the environment relevant to planning.
- **Actions (A):** The action space consists of discrete steering and speed commands, $(\delta_{\text{steer}}, \delta_v)$. Steering is discretized into *left*, *straight*, and *right*, and the speed change is discretized into *increase*, *maintain*, and *decrease*. In the LS variant, steering is determined by the preplanned path and the action space reduces to speed commands only.

- **Transition Function (T):** The transition function advances each component of the state by one time step. The target is propagated forward using the chosen motion model and is clipped to remain within the environment bounds. The vehicle’s pose is updated by applying the steering and speed commands under the vehicle’s kinematics over the time step, which is subdivided into finer segments for collision checking. Each nearby human moves toward its sampled goal with a small random perturbation to capture variability in human motion.
- **Observations (O):** The observation space consists of a discretized set of possible positions for nearby humans and the target. Velocities are not directly observed. In practice, the target’s state is estimated via a separate particle filter, which uses its own sensor readings and feeds a belief over the target’s state (position and velocity) into the planner, while the POMDP observation model captures the noisy positional information available to the robot.
- **Reward (R):** The reward function is designed to encourage safe, fast, and smooth motion. Its components include: (1) a large negative reward for collisions to promote safety; (2) a positive reward for successfully intercepting the target; (3) a small negative reward per time step to encourage shorter interception times; (4) a negative reward proportional to the deviation from maximum speed to promote fast progress; (5) a negative reward proportional to the magnitude of the steering angle to encourage smooth turning; and (6) an additional penalty for strong braking actions to discourage unnecessary emergency stops.

C. DESPOT

Determinized Sparse Partially Observable Tree (DESPOT) is an online POMDP solver that constructs a sparse search tree rooted at the current belief by sampling a finite set of scenarios. Each scenario represents one possible sequence of future states and observations, sampled from the POMDP’s initial belief and stochastic dynamics. Conditioned on a given scenario, future transitions and observations become deterministic, allowing DESPOT to “determinize” the problem along that scenario. By focusing on a limited number of scenarios, DESPOT efficiently approximates the value of actions without exhaustively exploring the belief space [6].

The algorithm maintains lower and upper bounds on the value function to guide the tree search and prune suboptimal branches, concentrating computation on the most promising actions. At each decision point, DESPOT selects the action that maximizes the estimated value over the sampled scenarios, using these bounds to balance exploration of new branches and exploitation of high-value ones.

In this project, the planner uses ARDESPOT (from POMDPs.jl) applied to the generative POMDP model defined above to select actions online. When the true target position is not directly observed, the scenarios are initialized using the most likely target estimate from the particle filter, which is

then propagated forward under the target motion model. The upper bound is computed using a fast heuristic that assumes driving straight toward the current most likely estimate of the target at maximum speed, ignoring obstacles, which yields an optimistic estimate of the achievable value. The lower bound is given by a reactive controller that slows down or brakes when humans are too close, and, for the ES planners, chooses a steering action that points the vehicle toward the predicted target. These heuristic policies define the lower- and upper-bound estimates and provide rollout values at the leaf nodes of the DESPOT tree.

D. Particle Filter

In this project, the target state is estimated using a bootstrap-style particle filter with several modifications tailored to the target-tracking problem. The filter maintains a set of N particles, each representing a hypothesis about the target’s state, including its position and velocity. Particles are propagated using one of two motion models: a linear constant-velocity model, or a nonlinear constant-speed, constant turn-rate model with boundary-induced heading changes. The observation model is either a finite-range omnidirectional sensor or a lidar-like sensor with finite range and limited field of view.

At each time step, particles are propagated forward through the chosen motion model with added process noise and boundary constraints. The weight of each particle is then updated according to the likelihood of the current observation given that particle’s state. When there is no detection, particles whose predicted positions lie inside the sensor’s field of view and range are downweighted, reflecting the negative information that the target was not observed where it would likely have been seen. Systematic resampling is performed, with a small jitter injected into the position and velocity, when the effective sample size falls below a threshold to avoid particle degeneracy.

When a detection occurs, the prior over the target’s position may be very diffuse, and the target could have moved almost anywhere, while the measurement provides a strong cue about its location. To avoid needing an impractically large number of particles to capture this posterior, the filter performs a measurement-driven rejuvenation step, in which particles are reinitialized in a tight cloud around the detected position and their weights are reset to be uniform. This can be interpreted as sampling from an importance distribution that approximates the posterior given the current measurement.

Velocity is not directly observed, but is inferred from finite differences of successive position measurements. Because this velocity estimate can be noisy, the filter does not directly overwrite the particle velocities with this estimate. Instead, it uses a Kalman-style blending:

$$v_{\text{new}} = (1 - \alpha) v_{\text{old}} + \alpha v_{\text{meas}}, \quad (4)$$

where $\alpha \in [0, 1]$ is a tunable blending factor. When observation noise is small, α is chosen close to 1, placing more weight on the measured velocity. When observation noise is large, α is chosen close to 0, relying more heavily on the prior velocity

estimate. Overall, this scheme yields a particle filter that is close to a bootstrap filter but augmented with measurement-driven proposals and adaptive velocity fusion.

IV. SOLUTION APPROACH

A. LS Planner Approach

Two variants of the Limited-Space (LS) planner are implemented in this work, with one variant serving as the baseline for evaluating the performance of the other planners.

1) *LS Baseline*: In this project, the LS baseline planner is used as the primary reference for comparison. At each decision step, a Hybrid A* path planner generates a kinematically feasible path from the current vehicle state to the Maximum A Posteriori (MAP) estimate of the target's position, given the current belief. If a new path cannot be computed within the allocated time budget, the planner falls back to reusing the previously computed path. Given the steering control sequence induced by this path, DESPOT then uses the path as a reference and optimizes only the speed of the vehicle, accounting for uncertainty in nearby human motion and avoiding collisions.

For computational tractability, the number of humans explicitly modeled in the POMDP solver is capped at six. When an additional (unmodeled) human comes within a specified safety radius of the vehicle, a sudden braking heuristic is triggered to avoid collisions.

In this approach, each human is represented as a static obstacle centered within a potential field whose spatial extent is related to the uncertainty over that human's intended goal. When a human's intention is highly uncertain, a larger potential field is placed around the human's current location; otherwise, the potential field is concentrated around the human's most likely path toward its goal. The path planner searches over 21 discrete steering actions uniformly spanning from -27.2° to 27.2° . The corresponding POMDP reward model is the same as described in Section III.B.

2) *LS-Intercept*: In the LS-Intercept variant, the Hybrid A* path planner generates a path to a *predicted future MAP estimate* of the target's position based on the current belief. First, the planner approximates a time-to-go to intercept the target using the nominal planning speed of the vehicle and the Euclidean distance to the target's current MAP estimate:

$$d = \|x^{\text{veh}} - x_{\text{MAP}}^{\text{tgt}}\|_2, \quad (5)$$

$$t_{\text{go}} = \frac{d}{v_{\text{plan}}}. \quad (6)$$

The target's MAP estimate is then propagated forward under the target motion model for duration t_{go} to obtain a predicted future position,

$$x_{\text{pred}}^{\text{tgt}} = \text{propagate}(x_{\text{MAP}}^{\text{tgt}}, t_{\text{go}}). \quad (7)$$

The predicted position is clamped to lie within the environment boundaries, and t_{go} is bounded to avoid unrealistically long prediction horizons. Hybrid A* then solves for a steering sequence $\{\delta_k\}$ that minimizes a cost combining collision penalties and path length.

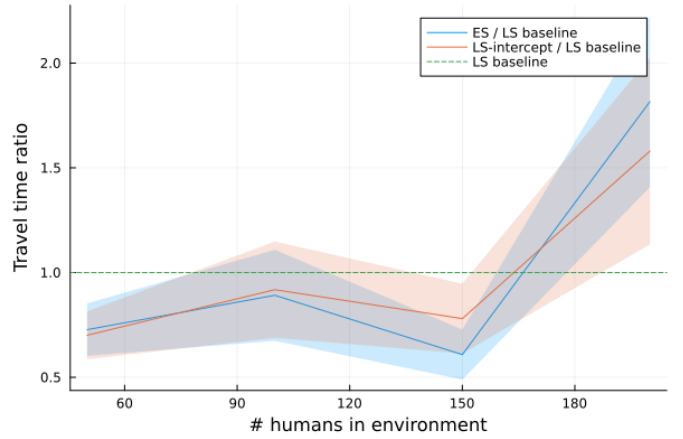


Fig. 2. Average interception time versus number of humans for each planner.

B. ES Planner Approach

At every decision step, the Extended-Space (ES) planner uses DESPOT to jointly control both the steering and speed of the vehicle. Since DESPOT is most effective with discrete, moderately sized action spaces, the continuous control space is discretized into a small set of actions (8 - 10 in number) that depends on the current vehicle state, in particular its speed v_a . Each action is a pair $(\delta_{\text{steer}}, \delta_v)$ consisting of a discrete steering command and a discrete change in speed.

The number and type of available actions vary with the current speed. When the vehicle is nearly stationary, the planner can choose to remain stopped or to move forward; at higher speeds, the actions correspond to moving forward with several discrete steering angles. The maximum steering magnitude is limited to $\pm 27.2^\circ$. The POMDP reward model is the same as described in Section III.B. At the leaf nodes of the DESPOT tree, a simple greedy rollout policy is used that repeatedly selects the action that moves the vehicle closer (in Euclidean distance) to the predicted most likely target position, and the resulting discounted return along this rollout is used as the value estimate for the corresponding leaf.

V. RESULTS AND DISCUSSION

The performance of the three planners described in Section IV is evaluated in a common simulated environment. We focus on three main aspects: (i) how often each planner successfully tags the target, (ii) how safely it does so (collisions and collision-free rate), and (iii) how efficiently it moves (interception time and sudden-brake usage).

A. Experimental Setup

All experiments use the environment shown in Fig. 1. The vehicle is equipped with a LiDAR sensor with a 15 m range and a 180° field of view, and its maximum speed is set to 2.0 m/s. The target moves at 1.65 m/s. Each planner is evaluated over 50 trials at four crowd densities: 50, 100, 150, and 200 humans. Each episode is capped at 180 s of simulated time.

TABLE I

RAW OUTCOME COUNTS AND AVERAGE EPISODE TIME OVER 50 TRIALS PER PLANNER AND CROWD LEVEL. #SUCC. = NUMBER OF RUNS THAT REACHED THE TARGET; #FAIL. = RUNS THAT DID NOT REACH THE TARGET; #COLL. RUNS = RUNS WITH AT LEAST ONE PEDESTRIAN COLLISION (SUCCESSSES OR FAILURES); MEAN EP. TIME (S) = AVERAGE EPISODE LENGTH OVER ALL RUNS, INCLUDING FAILURES.

# Humans	LS Baseline				LS-Intercept				ES			
	#Succ.	#Fail.	#Coll. runs	Mean ep. time (s)	#Succ.	#Fail.	#Coll. runs	Mean ep. time (s)	#Succ.	#Fail.	#Coll. runs	Mean ep. time (s)
50	40	10	14	105.69	46	4	8	64.96	48	2	0	58.19
100	28	22	23	120.52	39	11	12	82.34	47	3	0	60.64
150	22	28	13	140.43	30	20	17	116.23	48	2	0	55.29
200	20	30	24	132.60	28	22	25	111.62	38	12	1	94.48

TABLE II

AVERAGE TRAVEL TIME, NUMBER OF SEEDS WHERE A PLANNER OUTPERFORMED THE LS NON-INTERCEPT BASELINE IN TIME, AND AVERAGE NUMBER OF SUDDEN-BRAKE (SB) ACTIONS OVER 50 TRIALS IN ENVIRONMENT NO_OBSTACLES_50x50.

# Humans	LS Baseline			LS-Intercept			ES		
	Time (s)	# Outperf.	# SB	Time (s)	# Outperf.	# SB	Time (s)	# Outperf.	# SB
50	73.02 ± 8.81	–	3.94 ± 0.50	51.16 ± 5.58	12	2.70 ± 0.32	53.11 ± 6.53	19	2.60 ± 0.42
100	59.48 ± 12.69	–	4.20 ± 1.01	54.61 ± 7.11	7	3.38 ± 0.54	53.02 ± 6.22	12	2.38 ± 0.32
150	82.36 ± 13.51	–	4.67 ± 0.93	64.24 ± 8.78	3	3.91 ± 0.65	50.09 ± 5.38	12	3.06 ± 0.30
200	37.15 ± 7.00	–	2.08 ± 0.33	58.71 ± 12.30	1	3.71 ± 0.87	67.47 ± 8.27	5	4.21 ± 0.55

TABLE III

SUCCESS AND SAFETY STATISTICS OVER 50 TRIALS PER PLANNER AND CROWD LEVEL. SUCC. = FRACTION OF RUNS THAT REACHED THE TARGET; COLL.-FREE = FRACTION OF RUNS THAT REACHED THE TARGET WITHOUT RISKY SCENARIOS; MEAN COLL. = MEAN NUMBER OF DISTINCT COLLIDING PEDESTRIANS AMONG RUNS WITH AT LEAST ONE COLLISION.

# Humans	LS Baseline			LS-Intercept			ES		
	Succ. %	Coll.-free %	Mean coll.	Succ. %	Coll.-free %	Mean coll.	Succ. %	Coll.-free %	Mean coll.
50	80.0	62.0	1.57	92.0	80.0	1.12	96.0	96.0	0.00
100	56.0	40.0	1.48	78.0	64.0	1.42	94.0	94.0	0.00
150	44.0	36.0	1.54	60.0	46.0	1.29	96.0	96.0	0.00
200	40.0	26.0	1.54	56.0	34.0	1.44	76.0	76.0	1.00

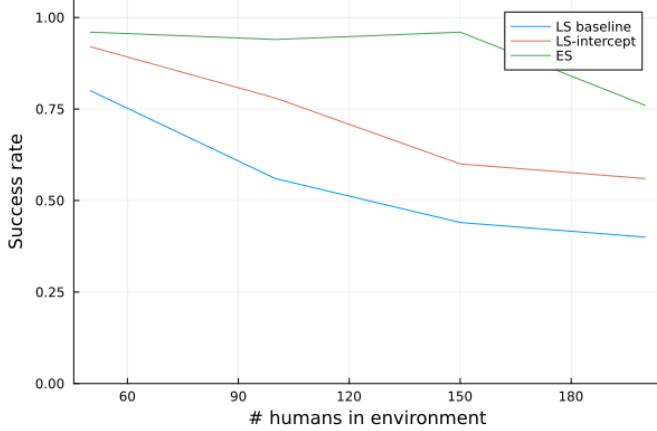


Fig. 3. Success and collision-free rates versus crowd density.

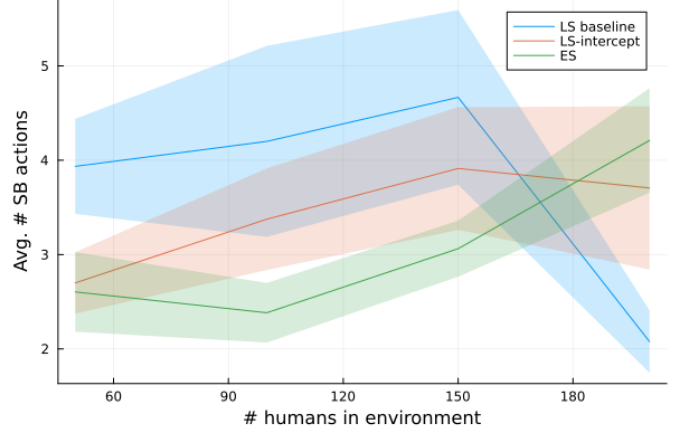


Fig. 4. Raw outcome counts and mean episode length across planners.

B. Raw Outcomes and Safety

Table I summarizes the raw outcome counts and mean episode lengths across crowd densities. The Extended-Space (ES) planner consistently achieves the highest number of successful runs and the fewest collision-involved runs in almost all crowd levels. The Limited-Space (LS) baseline is the most brittle; as the crowd becomes denser, its number of successes drops and the number of episodes with at least one pedestrian collision remains relatively high. The LS-Intercept planner

behaves as an intermediate option, it clearly improves on the LS baseline but does not match the ES planner.

These trends are reflected more compactly in Table III and Fig. 3, which show success and collision-free rates. The ES planner maintains a high success rate even at 200 humans and preserves a strong collision-free rate, whereas the LS baseline degrades sharply as density increases. LS-Intercept gains robustness over LS, but still underperforms ES at high densities.

C. Interception Time and Sudden Braking

Figure 2 and Table II report the average time to tag the target and the average number of sudden-brake (SB) actions, computed over successful runs for each planner and crowd level. For moderate crowd sizes (50–150 humans), ES and LS-Intercept both reduce interception time compared to the LS baseline, and ES typically uses fewer sudden-brake actions, indicating smoother and more anticipatory behavior.

At 200 humans, the mean interception times show a somewhat counterintuitive pattern: the LS baseline appears very fast on successful runs, while ES and LS-Intercept become slower and trigger more SB actions on average. This is largely a selection effect. When the crowd is very dense, the LS baseline succeeds only on relatively easy episodes, while many of the difficult trials fail due to timeout. Because the time and SB statistics only average over successful episodes, they exclude these hard failures and therefore paint an artificially optimistic picture for the LS baseline. In contrast, the ES planner succeeds on a larger set of episodes, including harder ones, which naturally increases its average interception time and SB count in this regime. The mean episode times over *all* runs in Table I give a more complete view and grow in a way that is more consistent with increasing crowd density.

An additional signal in Table II is the number of seeds on which a planner beats the LS baseline in interception time. In the 50-human setting, where all planners are relatively successful, ES outperforms the LS baseline in 19 out of 50 seeds, while LS-Intercept does so in 12 out of 50. This reinforces the picture that, whenever direct time comparisons are not dominated by success-selection effects, ES tends to reach the target faster than the LS variants.

D. Summary

Overall, the results indicate that the Extended-Space planner provides the best trade-off between success, safety, and efficiency. By jointly controlling both steering and speed, ES can thread through dense crowds more effectively than the Limited-Space planners, which are restricted to speed control along a fixed path. This additional control freedom enables ES to maintain higher success and collision-free rates while still keeping interception times competitive, especially in the more challenging high-density scenarios.

VI. CONCLUSION

This project presented a POMDP-based target-tracking and collision-avoidance framework for an autonomous vehicle operating in crowded human environments. By formulating the problem as a POMDP and comparing Limited-Space planners, which control only the vehicle’s speed along a precomputed path, against an Extended-Space planner, which jointly controls steering and speed, we showed that increasing the planner’s control authority can significantly improve performance. In our experiments, the Extended-Space planner achieved higher success rates, fewer collisions, and shorter interception times than the Limited-Space baselines, highlighting the benefits of online tree-search methods such as DESPOT

for handling long planning horizons in complex, uncertain environments. These results suggest that granting the POMDP planner greater freedom over the vehicle’s motion can yield safer and more efficient behavior when tracking a moving target in dense crowds.

We also observed that the quality of the rollout (heuristic) policy plays a crucial role in the performance of DESPOT, with reasonably designed rollouts enabling the solver to find good solutions across a range of scenarios. Future work could explore more informative rollout and bound heuristics, including learning-based proposals or value functions, to further reduce interception times and improve safety. In addition, integrating DESPOT with neural-network function approximators, for example, AlphaZero-style value and policy networks, would enable a systematic comparison against the current hand-crafted Extended-Space planner approaches.

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